- Many resource management problems may be framed as MAB, where we need to consider how to best allocate different resources to different organizations to best optimize some high-level objective (for example, profit or productivity) without knowing in advance how well each organization will operate. Portfolio management can also be framed as a MAB problem in the same way.
- MAB also has a long history in the design of clinical trials, where each patient needs to be assigned to a specific treatment. We wish to optimize the treatment outcome across all patients but need to content with limited resources and how likely each patient is able to benefit from a given treatment.

In each of these applications, we can take a set of actions, that is, a set of arms that we can pull, in order to optimize an objective under uncertainty.

5.1.2 From multi-armed bandit to Bayesian optimization

We have seen that MAB and Bayesian optimization (BayesOpt) have many shared characteristics. In both problems, we need to reason about what decision we should take so that we can maximize a quantity we care about. Further, the outcome of each action is not deterministic. This means that we don't know whether an action will yield a good result until we actually take it.

However, these two problems are not equivalent. In MAB, we aim to maximize the *cumulative reward* over time, that is, the total number of coins one receives. With BayesOpt, we seek to simply find an input of a function that leads to a high value, so as long as among the data set we collect there is a good objective value, we will succeed at optimization. This single-valued objective is sometimes called the *simple reward*. This difference means that we need to make sure we are frequently receiving rewards in MAB to maintain a good cumulative reward, while in BayesOpt we could afford to be more explorative to potentially find a good objective value.

SIDEBAR **Simple reward versus cumulative reward**

The term "simple reward" doesn't mean that the objective is easier or simpler to optimize, just that the objective is a single number, as opposed to a sum that is the cumulative reward.

Further, there are only a finite number of actions to take in MAB (a finite number of arms we could pull). In BayesOpt, as we are attempting to optimize an objective function in a continuous domain, there are infinitely many actions. Since we assume with a Gaussian process that function values at nearby points are similar to one another, we can think of this as actions that are close to each other yield similar reward rates. This is illustrated as Figure 5.4, where each infinitesimal point is a slot machine whose arm we could pull (that is, query the objective function value at), and machines that are close to each other have similar colors.

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